

MYSQL DATABASE PRACTICE

DDL Commands:

- SHOW databases; #show db
- DROP database bluepineapple; #drop db
- CREATE database bluepineapple; #it will create db name as bluepineapple
- USE bluepineapple; #(use) keyword for select db and work on it

Command for create Table:

- DROP table employee; # drop table if already present
- #command to create table
- CREATE table employee(eId int primary key, emp_name varchar(20), emp_qualification varchar(20), join_date Date, designation varchar(20), phone int);
- CREATE table department(dId int primary key, dept_name varchar(20), dept_head varchar(20));
- DESC employee; #it will shows the table structure

Field	Type	Null	Key	Default
eId	int	NO	PRI	NULL
emp_name	varchar(20)	YES		NULL
emp_qualification	varchar(20)	YES		NULL
join_date	date	YES		NULL
designation	varchar(20)	YES		NULL
phone	int	YES		NULL

- SELECT * from department; #it will shows the data from department table

dId	dept_name	dept_head
NULL	NULL	NULL

Alter Command:

- To add new column in existing table :
>>ALTER table employee add dId int references department ; #it will add 'dId' col
- To add Foreign key in existing table :
>>ALTER table employee add constraint fk_employee_dept foreign key(dId) references department (dId); #it make dId col as foreign key
>>DESC employee;

Field	Type	Null	Key	Default
eId	int	NO	PRI	NULL
emp_name	varchar(20)	YES		NULL
emp_qualification	varchar(20)	YES		NULL
join_date	date	YES		NULL
designation	varchar(20)	YES		NULL
phone	int	YES		NULL
dId	int	YES	MUL	NULL

- To Rename Column name :
 >>ALTER table employee rename column emp_qualification to qualification;
- To Rename Table Name :
 >>RENAME table department to departments;
- To Delete Table data without deleting entire structure using(TRUNCATE):
 >>truncate table employee;
- To Modify Data Type of column :
 >>alter table employee modify column eld varchar(20) ,
 modify column phone varchar(20) ;

DML Commands :

- **To Insert Values into the Table :**

>>INSERT into employee(eld, emp_name, qualification, join_date, designation, phone ,dId)
 values("Int004","Dhanesh", "MCA", "2025-12-15","Cloud Analyst",9356771276,"CId8748");

>>SELECT * from employee;

eId	emp_name	qualification	join_date	designation	phone	dId
Int004	Dhanesh	MCA	2025-12-15	Cloud Analyst	9356771276	CId8748
NULL	NULL	NULL	NULL	NULL	NULL	NULL

- **To Insert Multiple Values :**

>>insert into employee() values("Int001","Aniket","MSC","2025-07-18","Cloud Analyst",8754120396,"CId8748"),
 ("Int003","rohit","MSC","2025-07-18","DevCons",7845176041,"DevC7487"),
 ("Int013","yash","MCA","2025-12-18","DevCons",8947512463,"DevC7487"),
 ("Int008","Arin","MSC","2025-12-15","Tester",4578121010,"Test1015"),
 ("Int047","Roshan","MCA","2025-12-15","Data Analysts",5544552104,"DataA666"),
 ("Int019","Yuvraj","MSC Statistics","2025-07-18","DevCons",9191656582,"DevC7487"),
 ("Int012","Prerna","MSC Statistics","2025-12-15","Software Developers",6565894120,"Soft0001"),
 ("Int071","Arron","MSC","2025-12-18","Business Specialists",9798651201,"Buisn3333"),
 ("Int064","Pranav","MSC Data Science","2025-12-18","Project Leads",9784571240,"Pro5488");

>>SELECT * from employee;

eId	emp_name	qualification	join_date	designation	phone	dId
Int001	Aniket	MSC	2025-07-18	Cloud Analyst	8754120396	CId8748
Int003	rohit	MSC	2025-07-18	DevCons	7845176041	DevC7487
Int004	Dhanesh	MCA	2025-12-15	Cloud Analyst	9356771276	CId8748
Int008	Arin	MSC	2025-12-15	Tester	4578121010	Test1015
Int012	Prerna	MSC Statistics	2025-12-15	Software Developers	6565894120	Soft0001
Int013	yash	MCA	2025-12-18	DevCons	8947512463	DevC7487
Int019	Yuvraj	MSC Statistics	2025-07-18	DevCons	9191656582	DevC7487
Int047	Roshan	MCA	2025-12-15	Data Analysts	5544552104	DataA666
Int064	Pranav	MSC Data Science	2025-12-18	Project Leads	9784571240	Pro5488
Int071	Arron	MSC	2025-12-18	Business Specialists	9798651201	Buisn3333
NULL	NULL	NULL	NULL	NULL	NULL	NULL

- WE create Another Table name “departments” in that we insert multiple values:

```
>>INSERT into departments() values("Cld8748","Cloud","Natesh Vedgiri"),
```

```
("DevC7487","Devloper Consultancy","Jay Sheth"),
```

```
("Test1015","Testing","Avinash Thatte"),
```

```
("DataA666","Data Analytics","Suresh Burde"),
```

```
("Soft0001","Software Developer"," Raghu Chakravartula"),
```

```
("Buisn3333","Business Specialists"," S.M. Zia Alam"),
```

```
("Pro5488","Project Leading","Subhash Jadhav");
```

```
>>SELECT * from departments;
```

dId	dept_name	dept_head
Buisn3333	Business Specialists	S.M. Zia Alam
Cld8748	Cloud	Natesh Vedgiri
DataA666	Data Analytics	Suresh Burde
DevC7487	Devloper Consultancy	Jay Sheth
Pro5488	Project Leading	Subhash Jadhav
Soft0001	Software Developer	Raghu Chakravartula
Test1015	Testing	Avinash Thatte

- **Update Command :**

#It will update employee name Based on requirements.

```
>>update employee set emp_name = "Ankit" where eId = "Int001" and emp_name = "Aniket";
```

- **Delete Command :**

we add dummy record to perform delete command

```
>>INSERT into employee() values("Int081","Dummy","MSC","2025-12-18","Jr.Developer",9791111201,"Soft0001");
```

```
>>DELETE from employee where eId = "Int081"; #it will employee where id=int081
```

DQL Command:

```
>>select * from employee where qualification = "MCA"; #gives all emp who have MCA qualification
```

eId	emp_name	qualification	join_date	designation	phone	dId
Int004	Dhanesh	MCA	2025-12-15	Cloud Analyst	9356771276	Cld8748
Int013	yash	MCA	2025-12-18	DevCons	8947512463	DevC7487
Int047	Roshan	MCA	2025-12-15	Data Analysts	5544552104	DataA666

GROUP BY Command :

#it will group of designation give count of how many employee in each group

```
>>select designation ,count(eId) as Total_Employee
```

```
from employee
```

```
group by designation;
```

designation	Total_Employee
Cloud Analyst	2
DevCons	3
Tester	1
Software Developers	1
Data Analysts	1
Project Leads	1
Business Specialists	1

#Total count of employee

GROUP BY USING HAVING Command:

```
>>select designation , count(*)
from employee
group by designation
having designation ="Cloud Analyst";
```

designation	count(*)
Cloud Analyst	2

#it will group based on designation,After groping using having it will give only designation of certain group of count

WHERE CLAUSE WITH GROUP BY AND HAVING :

```
>>select designation ,count(*) as MSC_Qualifications_Count
from employee
where qualification = "MSC"
group by designation;
```

designation	MSC_Qualifications_Count
Cloud Analyst	1
DevCons	1
Tester	1
Business Specialists	1

#first using where clause it will select emp based on qualification=MSC ,then after on that records it will perform group by then from it using having we select particular records.

ORDER BY AND LIMIT CLAUSE :

```
>>SELECT * from employee ORDER BY join_date LIMIT 3;
```

eId	emp_name	qualification	join_date	designation	phone	dId
Int001	Ankit	MSC	2025-07-18	Cloud Analyst	8754120396	CId8748
Int003	rohit	MSC	2025-07-18	DevCons	7845176041	DevC7487
Int019	Yuvraj	MSC Statistics	2025-07-18	DevCons	9191656582	DevC7487

sort records based on join_date and using limit shows 3 records

LIKE operator :

```
>>SELECT * from employee where emp_name like "A%";
```

eId	emp_name	qualification	join_date	designation	phone	dId
Int001	Ankit	MSC	2025-07-18	Cloud Analyst	8754120396	CId8748
Int008	Arin	MSC	2025-12-15	Tester	4578121010	Test1015
Int071	Arron	MSC	2025-12-18	Business Specialists	9798651201	Buisn3333

it will gives all records of employee who's name start with 'A' Char.

JOINS :

1.INNER JOIN :

```
>>SELECT e.emp_name, d.dept_name  
FROM employee as e  
INNER JOIN departments as d  
ON e.dId = d.dId ;
```

emp_name	dept_name
Arron	Business Specialists
Ankit	Cloud
Dhanesh	Cloud
Roshan	Data Analytics
rohit	Devloper Consultancy
yash	Devloper Consultancy
Yuvraj	Devloper Consultancy
Pranav	Project Leading
Prerna	Software Developer
Arin	Testing

#it will join two table and select employee name from employee table and dept_name from departments using inner join

2.RIGHT JOIN :

```
>>select e.emp_name,d.dept_name  
from employee as e  
right join departments as d  
on e.dId = d.dId;
```

emp_name	dept_name
Arron	Business Specialists
Ankit	Cloud
Dhanesh	Cloud
Roshan	Data Analytics
rohit	Devloper Consultancy
yash	Devloper Consultancy
Yuvraj	Devloper Consultancy
Pranav	Project Leading
Prerna	Software Developer
Arin	Testing

it will gives all records from RIGHT Table and Matching Records From Left

3.LEFT JOIN :

```
>>SELECT e.emp_name,e.designation,d.dept_head  
FROM employee as e  
LEFT join departments as d  
ON e.dId = d.dId;
```

it will gives all records from LEFT Table and Matching Records From RIGHT

emp_name	designation	dept_head
Ankit	Cloud Analyst	Natesh Vedgiri
rohit	DevCons	Jay Sheth
Dhanesh	Cloud Analyst	Natesh Vedgiri
Arin	Tester	Avinash Thatte
Prerna	Software Developers	Raghu Chakravartula
yash	DevCons	Jay Sheth
Yuvraj	DevCons	Jay Sheth
Roshan	Data Analysts	Suresh Burde
Pranav	Project Leads	Subhash Jadhav
Arron	Business Specialists	S.M. Zia Alam

4.FULL OUTER JOIN :

- MySQL does not directly support the FULL OUTER JOIN keyword. To achieve the same result, you must combine LEFT JOIN, RIGHT JOIN, and the UNION operator.

```
>>SELECT e.emp_name,d.dept_name,d.dept_head,e.join_date
FROM employee as e
LEFT JOIN departments as d
ON e.dId = d.dId
UNION ALL
SELECT e.emp_name,d.dept_name,d.dept_head,e.join_date
FROM departments as d
LEFT JOIN employee as e
ON e.dId = d.dId;
```

emp_name	dept_name	dept_head	join_date
Ankit	Cloud	Natesh Vedgiri	2025-07-18
rohit	Developer Consultancy	Jay Sheth	2025-07-18
Dhanesh	Cloud	Natesh Vedgiri	2025-12-15
Arin	Testing	Avinash Thatte	2025-12-15
Prerna	Software Developer	Raghu Chakravartula	2025-12-15
yash	Developer Consultancy	Jay Sheth	2025-12-18
Yuvraj	Developer Consultancy	Jay Sheth	2025-07-18
Roshan	Data Analytics	Suresh Burde	2025-12-15
Pranav	Project Leading	Subhash Jadhav	2025-12-18
Arron	Business Specialists	S.M. Zia Alam	2025-12-18
NULL	AI	Jacky Barbara	NULL
Arron	Business Specialists	S.M. Zia Alam	2025-12-18
Ankit	Cloud	Natesh Vedgiri	2025-07-18
Dhanesh	Cloud	Natesh Vedgiri	2025-12-15
Roshan	Data Analytics	Suresh Burde	2025-12-15
rohit	Developer Consultancy	Jay Sheth	2025-07-18
yash	Developer Consultancy	Jay Sheth	2025-12-18
Yuvraj	Developer Consultancy	Jay Sheth	2025-07-18
Pranav	Project Leading	Subhash Jadhav	2025-12-18
NULL	Sales	Ray Devops	NULL
Prerna	Software Developer	Raghu Chakravartula	2025-12-15
Arin	Testing	Avinash Thatte	2025-12-15

it will all Matching and Non-Matching records from both table